

THE U.S. EXPERIMENTAL PROGRAMME FOR FAST REACTOR PHYSICS

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THE major experimental efforts in the United States programme for fast reactor physics centre on basic cross-section measurements, critical experiments, experiments on operating reactors, and experiments related to safety.

1. Basic Cross-section Data

At this stage of development of fast reactor physics and for a number of years to come, the determination of basic cross-sections and the measurement of integral and critical assembly parameters form a complementary, rather than a duplicative basis for the prediction of the reactor neutronic behaviour.

The United States has aggressively sought the required neutron microscopic data. Cross-section measurements useful to the fast reactor programme are obtained both from programmes specifically charged with supplying this information and from many other cross-section measurement programmes which have their main incentives in other areas. The effort has been pursued in three areas: (a) the resonance region, (b) the intermediate energy region, and (c) the continuum region. The investigations encompass fast neutron capture and scattering and the fission phenomena. An attempt is here made to cite outstanding improvements in each of the above areas and to point out technological and physical advances of particular relevance to the cross-section programme with a prognosis as to what can be expected in the future.

The Resonance Region ($E_n < 10$ keV)

The increasing size of fast reactors and safety considerations have particularly stimulated studies of the resonance nature of neutron cross-sections.⁽¹⁾ The majority of the work has exploited intense pulsed neutron sources and

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time-of-flight techniques to achieve excellent sensitivity and definitions often limited by the physical phenomena.⁽²⁾ The measurements have been principally limited to several large accelerators⁽³⁾ but also include a nuclear explosive which releases several moles of neutrons in a fraction of a μsec .⁽⁴⁾ Contiguous with the development of intense pulsed sources have been improvements in detector technology and, particularly, digital information acquisition and processing.

Recent work has defined in detail the total resonance cross-sections of an appreciable portion of the periodic table extending from Na-U.⁽⁵⁾ These results will not greatly affect Doppler estimates but they do determine strength functions requisite to the calculation of fission product effects and other phenomena. Both the total and partial radiative capture have been measured for a number of elements including Nb, W, and Th.⁽⁶⁾ One technique employs large liquid scintillators which also have been used to determine the capture to fission ratio, α , throughout the resonance region.⁽⁷⁾

Accelerator and explosive sources have also been used to measure fission cross-sections, particularly those of the Pu isotopes.⁽⁸⁾ The intensities available with an explosive make it possible to study small and very active samples. Thermal values of $\bar{\nu}$ are stabilizing principally due to increased refinement of the standard $\bar{\nu}$ of Cf²⁵².⁽⁹⁾ A source of possible error in predictions of integral parameters (e.g. the prompt neutron lifetime) is the spectrum of fission neutrons at low energies (≈ 50 keV).⁽¹⁾ Recent measurements tentatively indicate a relatively larger neutron yield in this low energy region than previously believed.⁽¹⁰⁾

Analytical and conceptual progress has not kept abreast of experiment in the resonance region. This is due to a relatively limited application of computational tools and to uncertainties in the physical theory. The latter tends to indicate that some resonance interpretations are little more than parametric representations of the data.

Intermediate Energy Region (10 keV–1.5 MeV)

Experiment in the intermediate region has been characterized by improved use of established techniques rather than the introduction of new concepts. Using these conventional methods it has been an area of outstanding accomplishment.

Representative of recent progress is the greatly improved knowledge of both elastic and inelastic neutron scattering in the intermediate region over a large portion of the periodic table.⁽¹¹⁾ The experimental resolutions of ≈ 20 keV define the scattering energetics to within the definitions of most multi-group cross-section sets and the experimental sensitivities permit a quantitative measure of differential cross-sections of a few millibarns. Unfortunately, the fissile nuclei most important for reactors are deformed and

consist of an odd number of nucleons. This makes determination of inelastic scattering difficult and uncertain.

Capture in the intermediate region has been extensively studied but the experimental results are not in good agreement.⁽¹²⁾ This is true of the sensitive quantities such as capture in U^{238} . Some of the discrepancies are due to uncertain standards employed in the various measurements. Thus, the renormalization of the existing values using the recent accurate determinations of the $\text{B}^{10}(n, \alpha)$ and $\text{Li}^6(n, \alpha)$ cross-sections in the region 10–200 keV⁽¹³⁾ should bring the values into better agreement.

Fission studies in the intermediate region have resulted in well defined $\bar{\nu}$ of $\text{U}^{235, 233, 238}$; $\text{Pu}^{239, 240}$; and Th^{232} .⁽¹⁴⁾ In all instances the energy dependence is well described by two linear functions of incident energy. However, detailed investigations of $\bar{\nu}$ (U^{235}) do indicate detailed energy-dependent structure having a magnitude of a fraction of a per cent. Preliminary measurements using an explosive source promise good capture and fission cross-sections for very active isotopes. Using conventional methods, α for U^{235} has been carefully redetermined with results similar to previously reported values.⁽¹⁵⁾

The wealth of scattering data in the intermediate energy region provides a good foundation for the optical model and the average compound nucleus model of Hauser–Feshbach.⁽¹⁶⁾ Nuclear potentials have been defined and proved valid for calculation of neutron scattering over an extended mass energy range.^(11, 17) Indeed, it is not *a priori* evident that a cross-section should be directly measured. Possibly more desirable is a calculation based upon a suitable potential. Unfortunately such calculations are difficult when applied to deformed nuclei such as $\text{U}^{235, 238}$.⁽¹⁸⁾

Continuum Region ($E_n > 1.5$ MeV)

Nuclear processes induced by neutrons of $\gtrsim 5.0$ MeV can be described by a continuum model of the nucleus whose average temperature is usually derived from measurement of the emitted particle spectrum.⁽¹⁹⁾ A number of such measurements has defined both elastic and inelastic neutron scattering processes at 5.0 MeV and above. The most serious uncertainties are between 1.5 and ~ 3.0 MeV: the region between the discrete structure observed at intermediate energies and the lower limit of valid experimental and theoretical interpretations based on a nuclear temperature. Uncertainties in inelastic scattering in this area are contributing factors in present discrepancies between integral calculation and experiment.⁽¹⁾

Neutrons of 5.0 MeV and above produce compound nuclei that decay by many modes. As a result, elastic neutron scattering is essentially all shape scattering and can be uniquely calculated from the optical model.⁽²⁰⁾ The results of such calculations are in such good agreement with measured

elastic scattering distributions that it is often preferable to calculate rather than measure elastic cross-sections at continuum energies.

Data Utilization

Intensive efforts have been made to compile the wealth of experimental information. Compilations in printed and machine (magnetic tape) formats covering the regions $Z \leq 20$ and $Z \geq 90$ have recently been widely distributed.⁽²¹⁾ However, the evaluation of the basic data to arrive at "best" values lags behind both the fundamental measurements and applied usage in reactor design.

Summary and Prognosis

The U.S. programme in basic cross-section measurements is meeting fast reactor needs in the intermediate and, to a lesser extent, resonance region. This is in no small part due to intense pulsed neutron sources obtained with large accelerators. The future of the programme will increasingly rely on these machines and new and more accelerators should soon become available. The unique intensities available with nuclear explosives promise greatly improved and more detailed knowledge of the cross-sections of highly active materials. Scattering cross-sections in the intermediate energy region are increasingly well known, and capture processes, though at present uncertain, should become well defined as the results of recent standard measurements are employed in the experimental work. Above ~ 5.0 MeV, the cross-sections are relatively less important in the present context and are perhaps now sufficiently well known. However, cross-sections in the region between 1.5 and 5.0 MeV are very doubtful and this fact leads to uncertainties in reactor calculations. In both the intermediate and continuum regions, increasing reliance should be placed on calculations using proven nuclear models to provide the desired information.

2. Critical Experiments

Many, although not all, of the neutronic problems of large dilute fast reactors can be studied on critical assemblies. Therefore, the critical experiment programme is an essential part of the fast reactor development programme.

ZPR 3 (ANL-Idaho) and ZPR 6 and 9 (ANL-Illinois) are carrying out critical experiments in support of the U.S. fast power breeder reactor programme. The Atomic International ECEL facility, which allows the study of fast zones surrounded by thermal driving regions, has also been used in the programme, particularly for the measurement of the Doppler effect in

recent years. All of these facilities and the planned Argonne ZPPR, to be used for large scale plutonium assemblies, are highly flexible, horizontally loaded, split table machines.

In this paper we concentrate on critical assemblies investigated in the past several years. (For earlier assemblies on ZPR 3 see ref. (22). For a description of ZPR's 6 and 9 and some of the experimental results on these facilities see refs. (23–26).)

Table 1 summarizes the results of the recent experiments using plutonium fuel. Recent assemblies have concentrated, in addition to mockups of specific reactors, on three general problems: the Doppler effect, the sodium void effect, and on the systematics and validity of zoned studies.

Doppler Effect Experiments

Doppler measurements have been made in recent years in the ECEL facility at AI and in ZPR 3 and 6 at Argonne. The Doppler measurement programme at AI is described in refs. (27–30) and includes measurements of U^{235} and U^{238} in various spectra at different sample temperatures.

Two sequences of Doppler effect measurements have been made in ZPR 3, the first in U^{235} fuelled cores^(22, 31) and the second in plutonium fuelled cores.⁽³²⁾ In the plutonium cores the experiments were done by exchanging hot and cold samples in a reactor whose central zone corresponded to the composition of a large plutonium–uranium (7:1) monocarbide (Assy. 45A), and to a graphite-softened version of the same (Assy. 45). The Doppler samples studied consisted of 0.50 in. diameter pellets of oxide sintered to 0.70 theoretical density, and stacked to a height of 6 in. Enriched Pu^{239} , mixed U^{238} and Pu^{239} (7:1), depleted uranium, and enriched uranium were used. Two Doppler samples were mounted in a vacuum jacket symmetrically about the axial centre of the reactor. After suitable checks and corrections, the change in reactivity on heating or cooling with respect to a reference material was ascribed to the Doppler effect.

The resulting ZPR 3 reactivity changes⁽³²⁾ are given in Table 2. In explanation of some of the items in the table it should be noted that most measurements were made in a local region where the cold plates of U^{238} and Pu^{239} were moved away from the Doppler sample, purely scattering materials being interposed. In the case labelled (homog), the central zone composition was made as uniform as possible. The effect of the change was not large. In another case, B_4 (natural) C was placed around the Doppler element to deplete the low energy end of the spectrum. The fact that the Doppler effect for Pu^{239} increased with this change indicates that better data on the resonance parameters of Pu^{239} in the energy region of up to about 2 keV would be particularly desirable. The Pu^{239} result reported here is the average of two independent measurements which agreed within 20%.

Values for the fissile samples have been corrected for expansion, which produces an effect calculated to be comparable to the Doppler effect.

The Argonne Doppler measurements have since been continued and extended in U^{235} assemblies in ZPR 6, further exploiting the technique in its potential for high precision. The flexible nature of the technique allowed the study of different effects such as different compositions and different arrangements of materials to be studied with ease. These experiments could therefore

TABLE 2. DOPPLER EFFECT REACTIVITY CHANGES 500 TO 1100°K

Sample	Diameter in.	Assembly	Experi- mental $\Delta k/k$ ($\times 10^{-6}$)	Theory $\Delta k/k$ ($\times 10^{-6}$)
U^{238}	0.5	45	-6.39	-6.19
U^{238}	1.0	45	-26.8	-26.11
Dummy	0.5	45	-0.112	
U^{238}	0.5	45A	-3.68	-2.66
U^{238}/Pu^{239}	0.5	45A	-3.12	-1.99
7:1 Mixture				
Pu^{239}	0.5	45	+1.07	+12.95
Pu^{239}	0.5	45A	+0.25	+5.06
Pu^{239}	0.5	45A(homog)	+0.67	
Pu^{239}	0.5	45A(B ₄ C)	+1.73	+2.33
U^{235}	0.5	45A	+2.15	+2.90

allow the study of a number of sources of possible uncertainty in the interpretation of the measurements. In refs. (33-35) the results of a number of measurements of this type are given. The phenomena studied related to the effect of an isolation zone around the sample, the effect of restraint on expansion, size effect of sample, effect of having diluent interspersed with sample, effect of void region surrounding the sample. The measurements were made with various sample fuel enrichments as a function of sample temperature.

Sodium Void Effect

The sodium void effect on reactivity for large fast reactors is important from safety considerations. It has been measured in the Fermi reactor,⁽³⁶⁾ on assemblies constructed on ZPR 3,^(37, 38) and during the past 2 years there have been two large carbide assemblies on ZPR 6, which have been thoroughly studied.^(25, 26) The first assembly studied on ZPR 6 had a 610 litre uranium carbide core with a length-to-diameter ratio of approximately unity. The second assembly had a flat pancake core with a volume of about 950 litres and a length-to-diameter ratio of $\sim 1:3$. In both cores, a thorough mapping of the sodium void coefficient as a function of location has been accomplished. The mapping of the sodium void coefficient was accomplished by

obtaining sodium versus void measurements in central regions with different diameters, in regions along the core axis, in cylindrical rings and finally in a sector comprising one-eighth of the total of the core.

Experiments to study the heterogeneity effect on the sodium coefficient were done in a central region of the 950 litre core assembly. Included were:

- (a) different materials placed next to the sodium,
- (b) different width of sodium columns,
- (c) partial voids in a 1 in. wide sodium column, and
- (d) discontinuous or staggered void in the normal drawer loading.

As much as about 20% variation between the highest and lowest value for the sodium coefficient was obtained with the various configurations. From these results, it would appear necessary to use two-dimensional transport calculations which properly takes into account the material arrangement in the drawers. Otherwise an agreement between experimental and theoretical results better than about 10% must be considered fortuitous. It would appear evident that for larger dilute cores where the spectrum is more degraded, it is even more important to take into account the heterogeneity effects.

Zoned Studies

Critical experiments include the multizone type in which the central test region is surrounded by a driver region, perhaps with an intervening buffer region having compositions quite similar to that of the central zone. Such loadings are particularly useful in the study of plutonium fuelled systems, in that the amount of plutonium required is much reduced. Numerical studies indicate that with large systems, central regions can be made sufficiently large to yield good results for some types of measurements and still lead to very appreciable reductions of material of the composition of interest.

An 1100-litre core volume, four radial region reactor (Assembly 4Z)⁽³⁹⁾ has been studied on ZPR 6 to determine the extent to which physics parameters of a large full scale reactor may be predicted on the basis of measurements in a zoned system. This assembly was composed of central, buffer, driver, and reflector radial regions with a core height of 122 cm. In the axial direction the central, buffer, and driver regions extended the full core height and were followed by a 30 cm thick depleted uranium reflector. The composition and dimensions of the regions are given in Table 3.

The central region composition was chosen to be identical to that for the 2600-litre (Assembly 5) core currently being studied on ZPR 6. The composition of the buffer and driver regions were chosen so that there would be a good calculated matching of the real and adjoint fluxes between the zoned core and the full scale 2600-litre system.

Central core spectrum which the Bennett proton recoil spectrometer,⁽⁴⁰⁻⁴²⁾

fission ratio measurements throughout the central zone, reactivity worth measurements of different material including core materials in the central, buffer and driver regions, were some of the studies conducted at the centres of both the zoned and full scale assemblies. In addition, comparative Doppler coefficient measurements of U^{235} and U^{238} and sodium void coefficient studies were made in the two assemblies.

TABLE 3. ASSEMBLY NO. 4Z 4-REGION REACTOR PARAMETERS

Material	Central $r_0 = 41.5$	Buffer $r_0 = 47.4$	Driver $r_0 = 54.5$	Reflector $r_0 = 91.2$
	Volume fractions			
U^{235}	0.0319	0.00095	0.1428	0.00169
U^{238}	0.2213	0.4539	0.1119	0.8319
Na	0.4208	—	0.4208	—
C	0.1539	0.3601	0.1539	—
SS (304)	0.1483	0.0708	0.1483	0.0708

Axial length of core regions: 122 cm

Axial reflector thickness: 30 cm

Critical mass: 1101 kg U^{235}

Mixed Spectrum Critical Assembly

An experimental programme has been conducted in the mixed spectrum critical assembly (MSCA) at the Vallecitos Atomic Laboratory to measure integral and differential parameters in a dilute fast ceramic reactor spectrum.⁽⁴³⁾ The mixed spectrum superheat reactor (MSSR) concept^(44, 45) consists essentially of an annular boiling water reactor coupled to a central fast spectrum superheater. The transition between the thermal and fast neutron spectra occurs in a buffer region which effectively filters out the thermal neutrons before they enter the fast core. The fast core volume in the MSCA is 400 litres. Both calculation and experiment demonstrate that the prevailing neutron spectrum in the central region at the desired power split (30% in superheater), which was present for the measurements, is asymptotic and representative of a dilute fast ceramic reactor (having a calculated mean fission energy of 200 keV and about 8% fissions below 9.1 keV).

A 4 π proton recoil proportional chamber has been used to measure the neutron spectrum directly between 12.5 and 115 keV by the method developed by Bennett.^(41, 42) The measured spectrum was softer than that calculated using existing cross-section data. Both measured and calculated spectra have dips at 19 keV, the former being the larger. This may be caused in part by the larger scattering resonances in Ni^{58} at 17 keV and Ni^{60} at 15 keV.

There is an accompanying capture resonance at 17 keV and, since there is about ten times more nickel than U^{235} in the fast core, this result suggests that the present^(46, 47) nickel capture cross-section at the 17 keV resonance may be too low.

The neutron spectrum between 0.143 and 2950 eV has been measured by six resonance detectors, Lu^{176} , Au^{197} , W^{186} , La^{139} , Cd^{114} , and Na^{23} . A sandwich foil technique, analysed by Judd,⁽⁴⁸⁾ was employed to distinguish between resonance and non-resonance neutron captures. For materials having several significant resonances, a correction factor was applied for the less important higher energy resonances.⁽⁴³⁾ The measured spectrum was again significantly softer than that calculated.

Two basic methods,⁽⁴⁹⁾ activation and reactivity, for measuring the Doppler effect have been employed in the MSCA. The two techniques are complementary since each measures a different component of the Doppler effect. The reactivity measurement, to first order, determines exactly the reactivity effect per gramme at the selected location of the reactor that would occur in a reactor power excursion, whereas the activation measurement determines the exact change in neutron capture or fission. The measured activation ratios⁽⁵⁰⁾ at 790°F were found to be substantially greater than predicted. The fission ratios of $U^{233, 234, 235, 236}$ and Np^{237} have been measured using very small silicon diodes. The measurements confirm evidence from direct neutron spectrum, neutron lifetime, and activation and reactivity Doppler effect measurements that the calculated spectrum is too hard.

A series of integral experiments was specifically directed at the reactivity effects of fast core flooding. This is an important safety consideration in both steam and gas cooled fast reactors since not only do they have a potential flooding accident, but deliberate flooding is considered during the refuelling operation. The experiments and their supporting analyses have indicated⁽⁵¹⁾ that it is entirely feasible to use resonance poisons for flooding reactivity control. For a modest reactivity penalty in the unflooded state ($\sim 1\% \Delta k$) quite large potential reactivity increases upon flooding (greater than $15\% \Delta k$) can be held within a fraction of a per cent Δk .

The prompt neutron lifetime and reactivity worths of the fast control and safety bundles were measured using pulsed neutrons by the $k\beta/l$ method.⁽⁵²⁾ Excellent agreement was obtained with period measurements indicating that discrete eigenvalues were excited in the coupled cores of the MSCA. This condition exists since the prompt neutron lifetime is dominated by the thermal region but would not be expected in a far subcritical reflected system where a continuum of eigen values is thought to occur. Of final note, all the aforementioned measurements were performed at essentially the same location, and for the same reference core composition, and thus the integral reaction and reactivity data are consistent with the spectrum determination—an important fact in the evaluation of fast cross-section data.

3. The SEFOR Reactor Programme

The SEFOR project, sponsored jointly by Southwest Atomic Energy Associates (a group of U.S. electrical utilities), Karlsruhe Nuclear Research Centre of West Germany, Euratom, the U.S. Atomic Energy Commission, and General Electric, is intended to obtain physics and safety information on an operating fast ceramic fuelled reactor. Particular emphasis is to be paid to the measurement and analysis of the Doppler effect under steady state and transient conditions.

The 20 MW(th) SEFOR reactor is sodium cooled and utilizes uranium-plutonium oxide fuel containing approximately 20% plutonium. The core is in the shape of a cylinder, approximately 34 in. in diameter and 34 in. high. The fuel pellet diameter in an individual pin is approximately 0.9 in., contained in a cladding of 0.040 in. stainless steel. The diameter of the fuel rods is made large so that at the relatively low SEFOR power density, the temperature distribution within the rod is very close to that expected within the fuel of a large fast ceramic breeder reactor. Similarly, the core composition is chosen so that the neutron spectrum and the Doppler coefficient are quite close to that expected in a large fast breeder power reactor. This is accomplished primarily by utilizing a high volume fraction of the core for fuel. Over 45 vol.% of the SEFOR core contains fuel. Studies by both General Electric and the Karlsruhe Nuclear Research Centre have shown that a low power experimental reactor with the SEFOR characteristics can give information on power coefficients directly applicable to high power reactors with small diameter fuel rods under both steady state and fast transient conditions.⁽⁵³⁾

The planned test programme for SEFOR has been divided into four parts. After the usual initial critical measurements, a series of steady state tests will take place. These tests involve the measurement of the change of reactivity between two different reactor operating conditions. For example, one test involves operation of the reactor at 10 MW(th) at half of full flow rate and then operation of the reactor at 20 MW(th) under full flow conditions. Since the sodium temperature remains essentially unchanged between these two conditions, the reactivity differences can be attributed (with small corrections) to the change in fuel temperature. Similar tests permit one to isolate the sodium and structure temperature coefficients, the flow coefficient, etc.

The second set of measurements are oscillator tests involving the oscillation of a special control element. At low frequencies (several cycles per minute or less) the Doppler feedback depends upon the Doppler coefficient divided by the fuel conductivity. These measurements will provide information similar to that obtained from the steady state tests, and thus provide a check on these tests. At high frequencies, i.e. from 10 c/min to 1 c/s,

the feedback response depends upon the Doppler coefficient divided by the heat capacity of the fuel. This is the same quantity which enters into the transient behaviour of the reactor. Thus, from the high frequency oscillator test, one can predict the results of the transient tests.

The transient tests are divided into two parts: sub-prompt critical tests and prompt critical tests. In the sub-prompt critical tests, reactivity insertions below β are involved. It can be shown that the transient response of the reactor can be analysed to obtain both the Doppler coefficient and the reactivity inserted. It is not necessary to know the reactivity which was inserted to analyse the results; alternately, if the reactivity inserted (as will be the case) is known, this provides a check on the analysis. Results of this test again involve the Doppler coefficient divided by the heat capacity and the analysis is made in terms of the delayed neutron parameters.

Following completion of the sub-prompt critical tests, a series of above prompt critical excursion tests will be performed. In this case, the analysis involves the prompt neutron lifetime and not the parameters of the individual delayed neutron precursors.

The various tests are complementary in the sense that they examine the Doppler effect under a variety of experimental conditions and their analyses involve the use of different reactor parameters. These tests, together with observations of the normal operating characteristics of the reactor, are intended to demonstrate that a fast ceramic reactor can be designed with desirable operating and safety characteristics. The information from SEFOR is expected to be of value in setting fast reactor safety design criteria, and in supporting such criteria during the various safeguards analyses which are part of the reactor licensing procedure.

At the present time, the SEFOR reactor is under construction at a site near Fayetteville, Arkansas. The reactor containment has been completed and internal structural work is in progress. The plant components, i.e. heat-exchangers, pumps, reactor vessel, etc., are being fabricated. An associated preoperational R&D programme involving fuel testing and tests on key elements of the plant is expected to be completed this year. Results to date, including initial tests in the ZPR 3 critical experiment and results of fuel tests, are encouraging and indicate that the goals of the programme can be met with the plant as designed. Initial dry critical testing of SEFOR is expected to take place at the end of 1967. More complete descriptions of SEFOR are given in refs. (53-56).

4. Experiments Related to Safety

Investigations into the phenomena and mechanisms of core meltdown accidents form the nucleus of the experimental fast reactor safety programme. Analysis of a meltdown incident requires knowledge of the causes of fuel

failure and movement, the modes of failure and material movement, rates of movement, and degrees of coherence in movement. Many complex and little understood phenomena are involved, including fuel slumping, cladding dissolution by molten fuel, transient stresses, fission gas effects, interactions between pins in clusters, and coolant boiling. For reactor designs with appreciable sodium void reactivity effects, coolant expulsion must be understood for its effects on reactivity as well as on fuel movement.⁽⁵⁷⁾

The experimental programme is centred on the transient reactor test facility (TREAT), which is being used to identify phenomena of interest, study their significance and interrelationships under in-pile conditions, and provide tests of theoretical predictions. Out-of-pile experimentation is being relied upon for model tests, basic data, and extension of in-pile investigations. The basic philosophy is to separate the various problems, such as pre-failure fuel motions, and fission gas effects, and study them as systematically as possible.

A key part of the programme consists of theoretical analysis of relevant phenomena, and experimental results, and of calculations relating results to reactor safety analyses.

Out-of-pile Experiments

Out-of-pile experimentation includes a continuing investigation of cladding penetration by fuel. First studies were related to the first core loading of EBR-II;⁽⁵⁸⁾ data on plutonium-bearing fuels are now being obtained.

Studies are also being made on the physical and chemical interactions resulting from sudden immersion in sodium of molten fuel materials. A broad range of heat transfer experiments is in preparation, including transient expulsion from heated sections, critical flow phenomena under accident conditions, superheat, transient heat transfer from small spheres moving rapidly in liquid sodium, and boiling.

Work is being initiated on filling in the most serious gaps in equation of state data, guided by theoretical evaluations⁽⁵⁹⁾ of the available information.

Meltdown tests were performed using electrically heated Fermi Core A type pins, both unirradiated⁽⁶⁰⁾ and irradiated.⁽⁶¹⁾ No stored energy effects were found that could accelerate fuel faster than gravity. Volume of the meltdown residue from irradiated pins was as much as three times the original volume.

Out-of-pile model tests were performed using four-pin clusters of electrically heated lead pins in flowing water as a simulation of Fermi Core A uranium alloy pins in sodium.⁽⁶²⁾ In most tests, initial movement of the simulated fuel was away from the core midplane. Subsequent flow may reverse this movement, but not produce velocities higher than predicted by a model of gross slumping under gravity.

TREAT Experiments

For convenience, TREAT results will be summarized by principal fuel types studied, rather than accident conditions or accident phenomena.

Metallic fuels

(a) EBR-II Mark I. This type has received the most extensive study. Failure of pins in an inert gas atmosphere has been characterized for both failure resulting from cladding dissolution by molten fuel and from bursting due to internal pressures. Theoretical analyses were checked by experiments on pins clad with niobium and with tantalum.⁽⁶³⁾ In-pile dissolution rate data⁽⁶⁴⁾ were also obtained. Pins run in stagnant sodium capsules⁽⁶⁵⁾ and in sodium loops⁽⁶⁶⁾ demonstrated the same failure mechanisms as samples run on inert gas, but analysis of results required careful interpretation of effects resulting from internal temperature gradients.⁽⁶⁷⁾ Gross failure under transient conditions of previously irradiated samples appeared to have the same causes and essentially the same thresholds as unirradiated pins, and led to formation of meltdown residue with sufficiently low density to play a significant role in limiting possible core compaction resulting from post-failure fuel motion.⁽⁶⁸⁾

(b) Fermi Core A. Both unirradiated⁽⁶³⁾ and irradiated⁽⁶⁸⁾ Fermi pins have been run in an inert gas atmosphere. Gravity has been found to be the dominant mechanism producing fuel motion. A simple model was adopted to characterize fuel behaviour.⁽⁶³⁾

(c) EBR-II Type Th-U. For conditions studied thus far, this type was found to retain sufficient high temperature fuel integrity at the time of failure that high velocity post-failure fuel movements did not occur.⁽⁶⁹⁾

Ceramic fuels

(a) UO_2 . Gas-bonded, EBR-II size UO_2 samples run in inert gas environments displayed failure behaviour in reasonable agreement with predictions of a simple model. High temperature failures of tantalum clad samples produced fuel fragment velocities up to 3 m/s.⁽⁷⁰⁾ Large diameter gas bonded fuel elements, contained in an NaK annulus, displayed extensive damage short of actual fuel failure upon repeated transient exposure of fuel to its melting point. Incipient fuel failure for a single transient was identified with a maximum fuel temperature $\sim 3870^\circ\text{C}$.⁽⁷¹⁾

(b) Mixed oxide. Experiments performed on fuel of this type contained in stagnant NaK resulted in densification and movement of fuel without cladding damage when a portion of the fuel passed through the heat of fusion.⁽⁷²⁾

(c) Steel clad, sodium bonded hypostoichiometric UC. Samples of this type, run in stagnant Na, survived excursions in which the fuel reached its melting point.⁽⁷³⁾ A higher energy transient failed one specimen, resulting

in extensive disposal of the fuel apparently due to local pressures in the sodium bond.⁽⁷⁴⁾

(d) Nb clad, unbonded, nominally stoichiometric UC. Samples of UC fuel contained inside evacuated niobium alloy tubing were run in inert gas, with failures observed due to transient mechanical loading and also to simple melting.

(e) US. EBR-II size, gas bonded US fuel in refractory metal jackets demonstrated fuel slumping and selective melting of fuel constituents at maximum transient temperatures within about 600°C of the melting point. Mechanical damage similar to that observed for oxide samples was not observed.⁽⁷⁵⁾

Application of results

Detailed use was made of meltdown data in a theoretical study of the degrees of coherence in failure for a range of hypothetical meltdown accidents in an EBR-II-type reactor.⁽⁷⁶⁾ These data, when combined with power and coolant flow variations over the core, showed that total times for fuel movement over the core could be much longer than values estimated on the bases of the time scale for fuel movement of a single pin.

Calculations have been made to relate the carbide fuel dispersal observed in TREAT to meltdown accident conditions. Shutdown due to the dispersal was shown to be quite effective in terminating the incident studied before propagation of meltdown throughout the core was complete.⁽⁷⁷⁾

In the near future, it is planned to extend the meltdown programme to study coolant phenomena and fuel movement more extensively, including effects of clustering and fission gases.

Proceeding parallel to these efforts are investigations of the basic data necessary to adequately describe meltdown incidents, including heat transfer, physical properties, and containment capabilities of cladding and structural material.

References

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